PlantDeepSUMO

A Deep Learning-Based Webserver for Predicting Lysine SUMOylation Sites in Plants

Quick Start Guide

Webserver: https://jasonxu.shinyapps.io/PlantDeepSUMO/

Source Code: https://github.com/Jasonxu0109/PlantDeepSUMO

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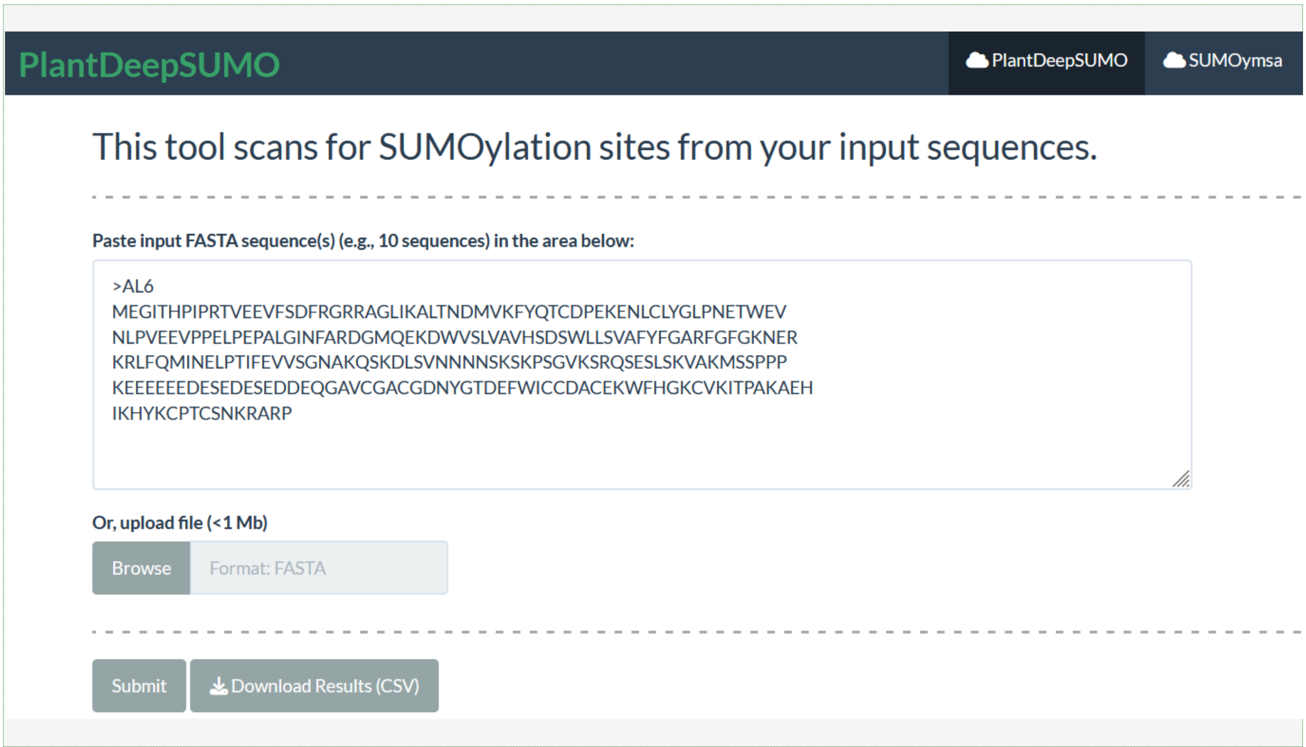
Overview

PlantDeepSUMO is a CNN-based deep learning tool that predicts lysine SUMOylation sites in plant proteins. It achieves an AUC-ROC of 0.93 and MCC of 0.75 on independent validation data, significantly outperforming existing tools. The webserver supports:

- Single & batch SUMOylation site prediction from protein sequences
- Interactive multiple sequence alignment viewer (SUMOymsa) for predicted sites

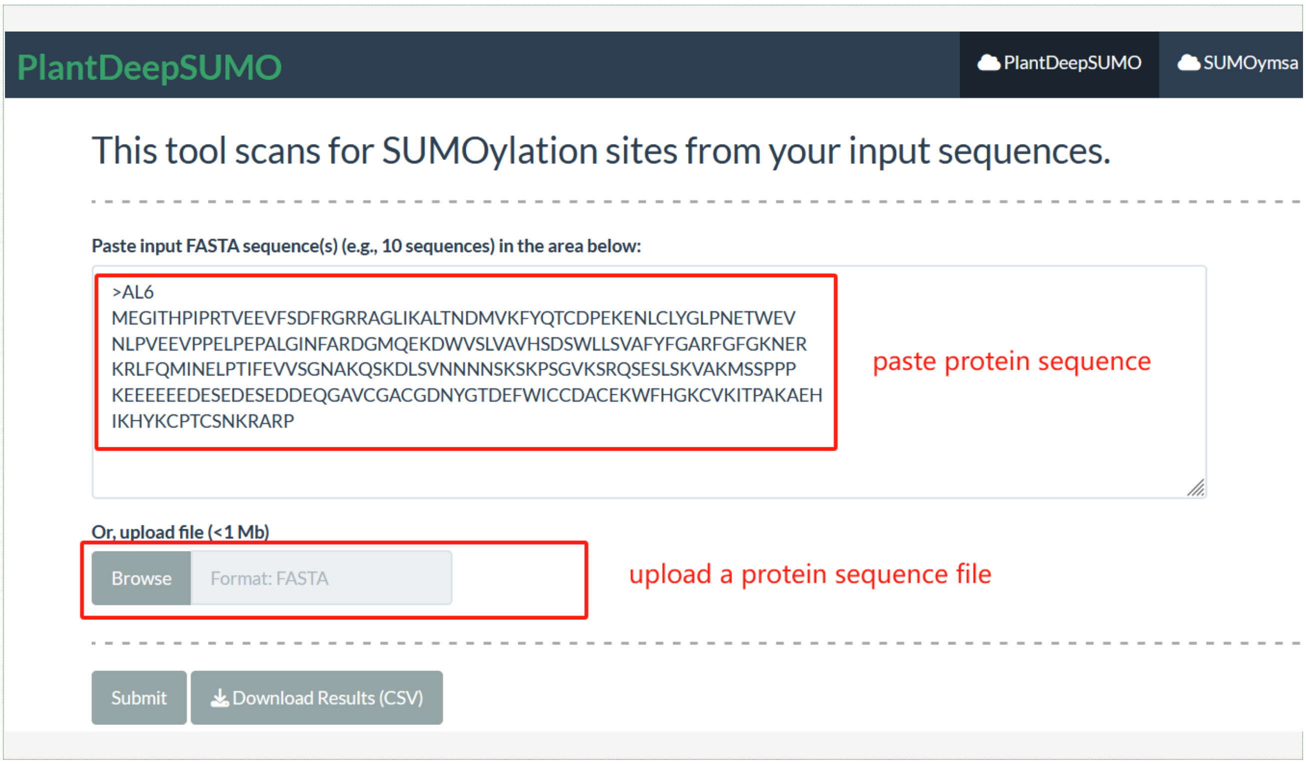
Quick Start: SUMOylation Site Prediction

Step 1: Open the Webserver



Navigate to https://jasonxu.shinyapps.io/PlantDeepSUMO/ in your web browser. The prediction interface will load automatically.

Step 2: Input Your Protein Sequence



You can provide protein sequences in FASTA format using either method:

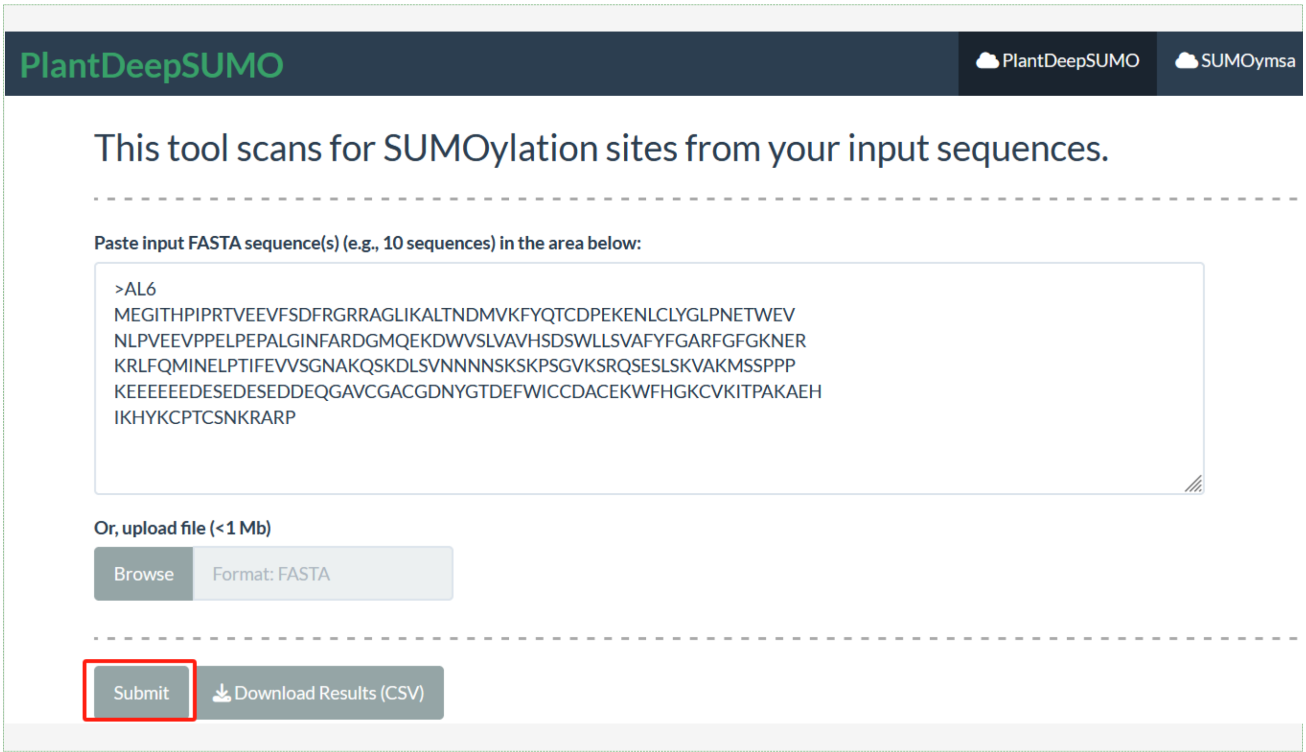
- **Paste directly:** Type or paste your FASTA-formatted sequence(s) into the input text box.
- **Upload a file:** Click the "Upload" button to select a FASTA file from your computer.

Example FASTA format:

```
>AT1G01010 | Arabidopsis thaliana protein
MDPNVVVDFSTANDKSVERRNVNKLTAYSKGKI...
```

Tip: For batch prediction, include multiple FASTA entries in one submission. The server handles files up to ~1 MB (~2,000 sequences).

Step 3: Submit and Wait for Results



Click the "Submit" button. Predictions for typical sequences are generated within seconds. For large batch submissions, please allow additional processing time.

Step 4: Review Prediction Results

Submit | Download Results (CSV)

Prediction of SUMOylation sites below:

Show 10 entries

Search:

	Name	Gene	Position	Probability	Confidence	Flanking_sequence
1	AL6:28	AL6	K28	0.921	very high	XXXMEGITHPIPRTVEEVFSDFRGRRAGLIALTNDMVKFYQTCDPEKENLCLYGLPNETW
2	AL6:36	AL6	K36	0.9731	very high	HPIPRTVEEVFSDFRGRRAGLIKALTNDMVFYQTCDPEKENLCLYGLPNETWEVNLPVEE
3	AL6:45	AL6	K45	0.7808	median	VFSDFRGRRAGLIKALTNDMVKFYQTCDPEENLCLYGLPNETWEVNLPVEEVPPELPEPA
4	AL6:88	AL6	K88	0.9247	very high	WEVNLPVEEVPPELPEPALGINFARDGMQEDWVSLVAVHSDSWLLSVAFYFGARFGFGKN
5	AL6:117	AL6	K117	0.9759	very high	EKDWVSLVAVHSDSWLLSVAFYFGARFGFGNERKRLFQMINELPTIFEVVSGNAKQSKDL
6	AL6:121	AL6	K121	0.9729	very high	VSLVAVHSDSWLLSVAFYFGARFGFGKNERRLFQMINELPTIFEVVSGNAKQSKDLSVNN
7	AL6:142	AL6	K142	0.8206	high	RFGFGKNERKRLFQMINELPTIFEVVSGNAQSKDLSVNNNNSKSKPSGVKSRQSESLSKV
8	AL6:145	AL6	K145	0.9722	very high	FGKNERKRLFQMINELPTIFEVVSGNAKQSDLSVNNNNSKSKPSGVKSRQSESLSKVAKM
9	AL6:155	AL6	K155	0.9276	very high	QMINELPTIFEVVSGNAKQSKDLSVNNNNSSKPSGVKSRQSESLSKVAKMSSPPPKEEEE
10	AL6:157	AL6	K157	0.8805	high	INELPTIFEVVSGNAKQSKDLSVNNNNSKSPSGVKSRQSESLSKVAKMSSPPPKEEEEEE

Showing 1 to 10 of 21 entries

Previous 1 2 3 Next

The results table displays:

- Gene/Protein identifier
- Lysine residue position of the predicted SUMOylation site
- Flanking sequence around the predicted site
- Prediction probability score (0–1)

Click the "Download" button above the table to export results as a CSV file for downstream analysis.

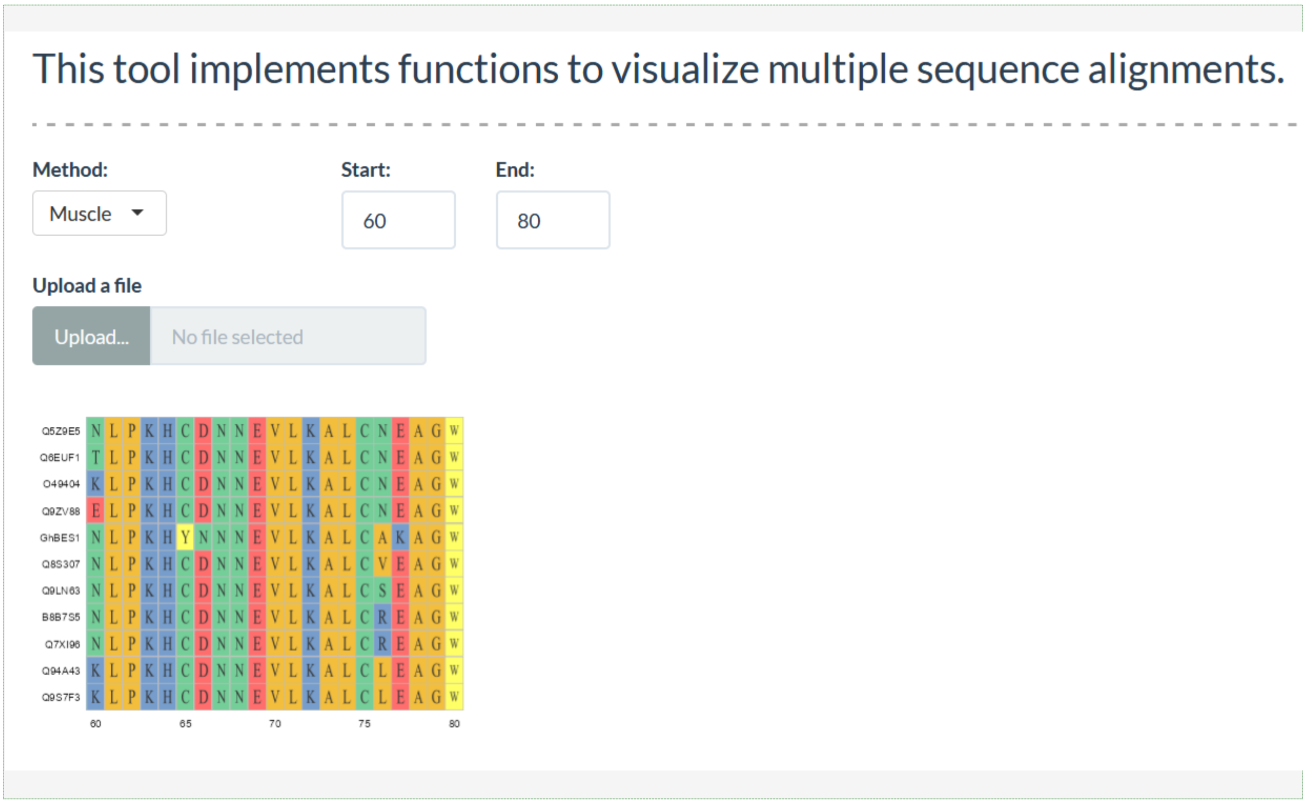
Interpreting Confidence Levels

Prediction probability scores are categorized into four confidence tiers:

Confidence Level	Probability Range	Interpretation
Very High	$0.90 \leq P < 1.00$	Strongly recommended for experimental validation
High	$0.80 \leq P < 0.90$	Reliable predictions; good candidates for validation
Medium	$0.65 \leq P < 0.80$	Moderate confidence; consider flanking context
Low	$0.50 \leq P < 0.65$	Lower confidence; use with caution

SUMOymsa: Sequence Alignment Viewer

The integrated SUMOymsa viewer allows you to visualize sequence conservation around any predicted SUMOylation site. After prediction, click on a specific site in the results table to open the alignment viewer and inspect flanking residue conservation across species.



Running Locally (Large Datasets)

For whole-proteome analyses or large-scale batch predictions beyond the web interface capacity, use the local execution scripts:

1. Download the Jupyter notebook from GitHub: example.ipynb
2. Install the required Python dependencies (TensorFlow/Keras, NumPy, etc.).
3. Run predictions offline using the pre-trained CNN model.

Citation

If you use PlantDeepSUMO in your research, please cite:

Xu, C. PlantDeepSUMO, a deep learning-based webserver to predict lysine SUMOylation sites in plants. (2026).